

OXIEDO: The Architecture That Unlocks Locked Data

The most valuable data in the world cannot be trained on.
ORMAS produces the account that
changes that, as the model learns it.

383 Controlled Experiments • Reviewer, NeurIPS 2026 Workshop • Submitted to AAAI 2027

| One Decision in 1986. Three Problems Everyone Funds Separately.

A Single Global Error Signal Makes Fault Localization Mathematically Impossible — Every Downstream Failure Follows From It

NOTHING CAN BE LOCATED

One error signal spreads across every parameter at once.

No unit can assess its own state.

The loss curve collapses and nothing says which node died.

NOTHING IS RETAINED

New tasks overwrite prior representations indiscriminately.

ResNet-18 keeps 47.3% after a sequential task shift.

Every regime change costs what the model knew.

NOTHING IS REPAIRED

Repair requires knowing what is broken.

Corrupted data is learned silently and confidently.

Rollback or full retrain are the only recourse.

One cause: the chain rule multiplies every unit's contribution through every other unit's. Unchanged since 1986.

| Kill a Converged Layer. One Architecture Comes Back.

Bounding Each Node's Local Gradient Chain Produces Attribution as a Physical Byproduct — and Attribution Makes Repair Possible

STANDARD BACKPROPAGATION

One global error signal, entangled across all parameters.

- No per-node visibility.
- Silent collapse on node death.
- No diagnosis, no targeted repair.

Feature layer zeroed at epoch 100:
10.0% — chance, every seed, zero variance, permanent.

ORMAS THREE-SIGNAL TRAINING

Each node reads through a **4,416-parameter** bottleneck — too small to memorise noise.

- Scores its own health every step.
- Fires at **2.5σ** below its own history.
- Diagnoses 7 modes, repairs mid-run.

Same attack: **80.3%** after 85 targeted corrections (**+70.3pp**).

Local stability characterization from Input-to-State Stability (Sontag, 1989) — not a global convergence proof, not yet peer-reviewed.

| The Industry Is Moving From Inference to Training on Private Data

Three Independent Pressures Converge on the Same Missing Layer — None of Them Existed Two Years Ago

REGULATORS SHIFTED TO ARCHITECTURE

The FDA rejects medical models for opacity **regardless of accuracy**.

The EU AI Act attaches liability to systems with no audit trail.

The requirement is no longer performance. It is provenance.

RENTED INFERENCE STOPPED SCALING

Per-call cost compounds linearly with usage.

At institutional volume the only exit is your own infrastructure.

Which means training on your own data — the thing nobody can do safely.

MODELS DEGRADE INVISIBLY

Every regime change, patient cohort and fraud pattern erodes what the model knew.

No detection exists at the training layer.

It surfaces as a miscalculation, not an alert.

The regime is shifting from inference on public data to training on proprietary institutional data.

| The Market, Counted From the Constraint Upward

Sized Bottom-Up From Institutions That Qualify — Not Down From an Analyst Report

TAM \$8.3B (2026)

AI model-risk management, worldwide.

\$19.1B by 2035, 9.7% CAGR.

The budget exists. Today it buys tools that estimate a finished model from outside.

SAM \$250M

~1,000 institutions that qualify on all three tests:

- ~180 academic medical centres with an AI group and a regulated pathway
- ~400 funds and asset managers with in-house ML
- ~270 insurers and credit-risk teams under supervisory reporting
- ~150 labs running training at cost

× \$250K ACV = \$250M

SOM \$5M ARR

Beachhead: regulated finance.

Shortest cycle, clearest mandate, a named validator on the other side, and no ethics board.

20 institutions by end of year 3 at \$250K.

2% of SAM. The constraint is delivery capacity, not demand.

TAM: SNS Insider, AI model-risk management, 2026. Institution counts are our own estimates, and no buyer interviews have happened — the number most likely to be wrong.

| One Product. Three Things Arrive With the Model.

Training Is Unchanged — Data, Objective and Resulting Model Are Identical. What Changes Is the Account That Exists Alongside It

01 — THE MODEL

Deploys exactly as it would have

Same architecture, same inference cost, no change to the serving path.

Nothing is traded away for the accountability. That is the whole point.

02 — THE NAMED PARTS

What it learned, per source

Each source's contribution held as a discrete addressable object, bound to the record of the data that justified it.

Removable on instruction, without retraining.

03 — THE RECORD

Every change training made

Which component, at which step, under which diagnosis, by how much, inside what declared limit.

Signed, and diffable against the last approved version.

Seven features of one architecture, not seven products. Nothing is priced per feature. [anonymous.4open.science/r/oramas-EB73](https://open.science/r/oramas-EB73) — all 383 runs reproduce with one command.

| The Evidence Is Open. Replay a Real Run in Your Browser.

Checkable Without Us in the Room — and One of Them Is Running Live Right Now

LIVE oxiedo.com/black-box

An archived 54-day training run, replayable epoch by epoch, two identical networks side by side.

- Step through epochs 0–199 and watch per-component health move before the loss curve does.
- Open the correction ledger: step, component, diagnosis, magnitude, declared ceiling.

Seeds are on screen. Every run regenerates.

BUILT A COMPANY, AS A CONTROL

Before the architecture, the same discipline applied commercially: a registered UK company run for twelve months as a controlled experiment.

Remove the system: -91% output, -100% new customers, every channel on the same day.

Put it back: $+1,300\%$, overshooting $3.3\times$ on memory alone.

What it demonstrates: experiments designed to fail in public, run to completion, reported either way.

The company experiment tested the orchestration layer and is prior work. The architecture carries its own record — 383 controlled experiments, 4 families, every run reproducible from seed, and one adverse result published alongside the rest.

| One Licence, Annual, Per Institution, On Their Metal.

Publish the Mechanism, License the Accountability — Priced Against Exposure Carried, Never Against Seats or Tokens

THE MODEL

Annual deployment licence, per institution, on-premise or in their own cloud.

No per-seat component. No consumption meter. No hosted tier, ever.

Target ACV **\$250K**, band \$150K–\$500K.

WHAT SETS THE FIGURE

Four inputs, written into the contract:

1. What the problem costs them today.
2. Model portfolio in scope.
3. Whether the bound exists or must be calibrated.
4. Liability scope accepted.

Assurance priced per engagement, on top.

WHY IT RENEWS

Artefact survival: on termination they keep every record, plus a frozen reader binary. Renewal is never hostage-taking.

Bound immutability: the declared bound is fixed at execution and moves only by signed amendment.

A validator who has accepted the method does not re-accept a replacement cheaply.

Gross margin is software-grade: nothing is hosted, so serving cost is the customer's. Zero contracts signed, zero customer conversations held — the first pricing conversation is the milestone, not the assumption.

| Go-To-Market: Three Motions, One Beachhead, Named Steps.

Regulated Finance First — Shortest Cycle, and a Named Validator on the Other Side

1 — TECHNICAL ENTRY

The falsification offer — under 30 GPU-hours, run by their own team.

Channel: direct outbound by name, the workshop and conference circuit, and the published preprint.

Turns a sales call into an experiment.

2 — REGULATORY ENTRY

The validator is the real gatekeeper, so engage them before the customer does.

Channel: model-risk bodies, supervisory consultations, and the Big Four validation practices already in these reviews.

A method a validator has seen is a shorter sale for every account after it.

3 — LAND AND EXPAND

Design partner (free, 3 months, read-only mode) → **one model in production** → **institution-wide licence**.

Read-only mode is the wedge: nobody lets an unproven system touch weights in a six-figure run, and it does not ask them to.

Cycle: 6–18 months.

Y1 3 design partners, \$0, transformer result published · **Y2** 4 paid licences, **\$1.0M ARR**, one validator acceptance · **Y3** 20 institutions, **\$5.0M ARR**, second sector opened.

Founder-led sales through years 1–2. The commercial hire lands against a pipeline, not ahead of one.

| Everyone Is Competing One Layer Above the Constraint

Orchestration Is Not the Blocker — the Blocker Is That You Cannot Safely Train on the Data

	Per-node attribution	Stability analysis	Autonomous repair	Audit trail at train time	Their metal
Agent frameworks	×	×	×	×	×
Enterprise data platforms	×	×	×	~	~
Frontier model vendors	×	×	×	×	×
Data labelling vendors	×	×	×	×	×
Academic research	~	~	~	×	—
This	✓	✓	✓	✓	✓

The moat is not secrecy — the work publishes on release. It is that getting here ran through a live company broken on its own data, then control theory, then two years of experiments. That order does not compress.

| The Company: What Is Built, What Is Bought, What Is Next.

One Person Produced the Architecture, the Evidence Base and the Commercial Proof — the Round Converts That Into an Organisation

WHAT IS ALREADY BUILT

Rokib Al Dhin Raadh — founder.

- The architecture, with a formal stability result, derived at 17.
- 383 controlled experiments, 4 families, reproducible from seed.
- Reviewer, **NeurIPS 2026** workshop programme committee.
- A UK entity formed and operated from abroad, unassisted.

WHAT THIS ROUND BUYS

Two hires, in this order:

1. A **research engineer** — removes key-person risk on the architecture itself. Month 1.
2. A **commercial lead** who has sold inside a fund, hospital or insurer — lands in year 2, against a pipeline rather than ahead of one.

Founder-led sales until then, by design: the first conversations are technical.

WHAT IT IS NOT

Not a team looking for an idea. The architecture is finished and published.

Not a services business. One licence, one contract, no bespoke builds.

Not a hosted platform. Nothing leaves the customer's network, ever.

Not dependent on a frontier lab. No model rented, no API in the path.

Key-person risk is total today and it is the first line of the use of funds. raadh.me · oxiedo.com · zenodo.org/records/21730363

Financials: Burn, Runway, and When Revenue Starts.

Pre-Revenue by Design — Nothing Ships Until the Architecture Clears Its Own Destruction Suite

	Year 1	Year 2	Year 3	Assumption
Paying institutions	0	4	20	Land at 1 model, expand
ARR	\$0	\$1.0M	\$5.0M	\$250K ACV, flat
Gross margin	—	~92%	~92%	On-premise; no hosting cost
Net burn	\$820K	\$680K	Near breakeven	Compute front-loaded to year 1
Cash position	\$680K	Seed raised	Default alive	18 months on this round

Assumptions, so they can be argued with: \$250K ACV is benchmarked against model-risk tooling and **validated by no one — zero pricing conversations have happened.** Year 2 assumes the transformer result lands and one validator accepts the method; if either fails, year 2 is 1–2 design partners and a seed raised on research rather than revenue.

Seed assumed at month 15 against milestones, not ARR. This round does not reach default-alive and is not shown as though it does.

| From a Training Layer to the Substrate Regulated AI Runs On

Every Phase Removes One Dependency — Ending With a Stack That Rents Intelligence From No One

PHASE 1 — NOW

Finish the research and carry per-node health monitoring onto attention heads.

Architecture-agnostic by design, so this is a compute problem, not a research one.

First design partners.

PHASE 2 — 12-24 MONTHS

First licensed deployments inside institutions that cannot move their data.

Each accumulates institutional memory that cannot be exported.

Switching cost compounds with tenure.

PHASE 3 — YEARS 3-5

A model pre-trained on the self-correcting architecture from scratch.

The external dependency disappears; cost collapses to hardware.

A position no competitor reaches by adding a feature.

The network learns to stop needing correction. That is the thesis, and it is measured.

Correction frequency decays $4.2 \rightarrow 0.05$ per epoch — the system repairs itself until it no longer needs to.

| \$1.5M Pre-Seed on a SAFE. Sized to the Next Result.

Eighteen Months, Six Milestones With Acceptance Criteria — the Compute Line Costed Against the Question Actually Open

THE INSTRUMENT

\$1.5M on a post-money SAFE.

\$12M cap, 20% discount, no debt component.

Implied dilution at conversion:
~12.5%.

Priced round deferred to the seed.

USE OF FUNDS

\$650K Research engineer; co-founder at below-market cash, equity-weighted

\$250K 32-64 accelerators, two campaigns

\$150K One custodian, non-exclusive

\$100K Delaware, EU rep, filings

\$200K 18 months of everything else

\$150K Buffer, unallocated on purpose

WHAT IT BUYS

1. A second and third person.

Key-person risk is total today.

2. One result that is not CIFAR.

Bounded attribution on a transformer backbone.

3. Three design partners and the first pricing conversation this company has ever had.

The frontier-scale campaign is a seed problem, and is not being asked for here.

Before writing a cheque, spend \$50 instead: the remaining programme is falsifiable in under 30 GPU-hours, on your hardware, without us in the room.